

Dominick Reilly

✉ dreilly1@charlotte.edu 🏠 dominickrei.github.io 🌐 dominickrei

Education

University of North Carolina at Charlotte

Aug 2022 - Present

Ph.D. Student in Computer Science

Advisor: Dr. Srijan Das

University of North Carolina at Charlotte

Jan 2019 - May 2022

Bachelor of Science in Computer Science, GPA: 4.0

Experience

University of North Carolina at Charlotte

Aug 2022 - Present

Research Assistant - Charlotte Machine Learning Lab

- Video understanding in large vision-language models using cross-perspective data (1st and 3rd person) for exocentric video understanding of daily living activities. [3]
- Multi-modal (RGB + Human Pose + Human Object Interactions) Large Vision Language Model (LVLM) for understanding daily living actions from video. Curriculum-based progressive training for integrating multiple modalities into the LVLM embedding space. [1]
- Multi-modal (RGB + Human Pose) and viewpoint agnostic video transformers for understanding subtle appearance and motion cues in human activities. [8]
- Self-supervised learning and masked autoencoders for training ViTs on small, out-of-domain data distributions. [9]

Honda Research Institute, California

Jun 2024 - Sep 2024

Student Researcher - Perception Team

- Training vision-language models on procedural activity datasets containing multiple perspectives (egocentric & exocentric), and transferring them to single-perspective datasets.
- Learning pose-augmented RGB representations in CNNs for weakly-supervised action and error detection in procedural activity videos. [4]

Inria, Sophia Antipolis, France

Jan 2024 - Jun 2024

Research Intern - STARS Team (Chateaubriand Fellow)

- Incorporating the human pose modality into vision-language models (CLIP) for better zero-shot understanding of daily living actions from video. [6] [2]
- Improving temporal action detection with gating mechanisms to improve feature selection. [7]

University of North Carolina at Charlotte

Jul 2021 - Jul 2022

Research Assistant - Data Privacy Lab

- Safeguarding face and iris images from deep-learning based re-identification attacks. [11]
- Created interactive webpage demonstrating safeguards on face and iris images. Try the demo for yourself at <http://3.223.148.187/>. [10]

Publications (C)onference, (P)reprint, (W)orkshop

1. (C) **Dominick Reilly**, Rajat Subhro Chakraborty, Arkaprava Sinha, Manish Kumar Govind, Pu Wang, Francois Bremond, Le Xue, Srijan Das, “LLAVIDAL: A Large Language Vision Model for Daily Activities of Living,” IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2025.
2. (C) Arkaprava Sinha, **Dominick Reilly**, Francois Bremond, Pu Wang, Srijan Das, “SKI Models: Skeleton Induced Vision-Language Embeddings for Understanding Activities of Daily Living,” Proceedings of the AAAI Conference on Artificial Intelligence (AAAI), 2025.
3. (P) **Dominick Reilly**, Manish Kumar Govind, Srijan Das, “From My View to Yours: Ego-Augmented Learning in Large Vision Language Models for Understanding Exocentric Daily Living Activities,” arXiv:2501.05711, 2024.
4. (P) Seth Z. Zhao, Reza Ghoddoosian, **Dominick Reilly**, Isht Dwivedi, Nakul Agarwal, Behzad Dariush, “Pose-Aware Weakly-Supervised Action Segmentation,” , 2024.
5. (P) Ali Khaleghi Rahimian, Manish Kumar Govind, Subhajit Maity, **Dominick Reilly**, Christian Kümmerle, Srijan Das, Aritra Dutta, “Fibottention: Inceptive Visual Representation Learning with Diverse Attention Across Heads,” arXiv:2406.19391, 2024.
6. (W) Mahmoud Ali, Di Yang, Arkaprava Sinha, **Dominick Reilly**, Srijan Das, Gianpiero Francesca, Francois Bremond, “Quo Vadis, Video Understanding with Vision-Language Foundation Models?,” Workshop on Video-Language Models (NeurIPS), 2024.
7. (W) Aglind Reka, Diana Laura Borza, **Dominick Reilly**, Michal Balazsia, Francois Bremond, “Introducing Gating and Context into Temporal Action Detection,” Workshop on Affective Behavior Analysis in-the-wild at (ECCV), 2024.
8. (C) **Dominick Reilly**, Srijan Das, “Just Add π ! Pose Induced Video Transformers for Understanding Activities of Daily Living,” IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024.
9. (C) Srijan Das, Tanmay Jain, **Dominick Reilly**, Pranav Balaji, Soumyajit Karmakar, Shyam Marjit, Xiang Li, Abhijit Das, Michael Ryoo, “Limited Data, Unlimited Potential: A Study on ViTs Augmented by Masked Autoencoders,” IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2024.
10. (C) Muhammad Saleem, **Dominick Reilly**, Liyue Fan, “DP-Shield: Face Obfuscation with Differential Privacy,” International Conference on Extending Database Technology, 2022.
11. (C) **Dominick Reilly**, Liyue Fan, “Comparative Evaluation for Differentially Private Image Obfuscation,” IEEE International Conference on Trust, Privacy and Security in Intelligent Systems, and Applications (IEEE TPS), 2021.

Academic Activities

- Reviewer for ICCV 2025, CVPR 2025, AAAI 2023/24/25

Awards

1. The Chateaubriand Fellowship (awarded by the Embassy of France), 2023
2. Best poster award in Mathematics and Computer Science, UNC Charlotte Undergraduate Research Conference, 2020